Teorema de la derivada cruzada

$$\frac{\partial^2 f(x,y)}{\partial x\,\partial y} = \frac{\partial^2 f(x)}{\partial y\,\partial x}$$

escalar

vector

b. $\nabla\cdot(\nabla\times\mathbf{u})=0$

$$\nabla\cdot(\underbrace{\nabla\times u}_{v}) = \frac{\partial}{\partial x_i}(v_i) = \frac{\partial}{\partial x_i}\left(\varepsilon_{ijk}\,\frac{\partial u_k}{\partial x_j}\right) = \varepsilon_{ijk}\,\frac{\partial^2 u_k}{\partial x_i\,\partial x_j}$$

constante

$$= \frac{1}{2}\left(\varepsilon_{ijk}\,\frac{\partial^2 u_k}{\partial x_i\,\partial x_j} + \varepsilon_{ijk}\,\frac{\partial^2 u_k}{\partial x_i\,\partial x_j}\right)$$

$$= \frac{1}{2}\left(\varepsilon_{ijk}\,\frac{\partial^2 u_k}{\partial x_i\,\partial x_j} - \varepsilon_{jik}\,\frac{\partial^2 u_k}{\partial x_j\,\partial x_i}\right) = 0$$

$$\varepsilon_{ijk}\,u_j\,u_k$$

$$-\varepsilon_{ikj}\,u_k\,u_j$$

escalar

c. $\nabla\times(\nabla\phi)=0$

$$\underbrace{\nabla\times(\underbrace{\nabla\phi}_{v_K})}_{u_i}$$

$$(\nabla\times\nabla\phi)_i = \varepsilon_{ijk}\,\frac{\partial}{\partial x_j}(\nabla\phi)_k = \varepsilon_{ijk}\,\frac{\partial}{\partial x_j}\,\frac{\partial\phi}{\partial x_k} = \varepsilon_{ijk}\,\frac{\partial^2\phi}{\partial x_j\,\partial x_k} = 0$$

$$\boxed{\nabla\times\nabla\phi = 0}$$

$$(\nabla\cdot T)_i = \frac{\partial T_{ij}}{\partial x_j}$$

$$(\nabla\cdot T^T)_i = \frac{\partial}{\partial x_j}(T^T)_{ij} = \frac{\partial}{\partial x_j}T_{ji}$$

d. $\nabla\times(\nabla\times\mathbf{u})=\nabla(\nabla\cdot\mathbf{u})-\Delta\mathbf{u}$

$\nabla\times\ \ v$

P

$$[\nabla\times(\nabla\times u)]_i = \varepsilon_{ijk}\,\frac{\partial}{\partial x_j}(\nabla\times u)_k = \varepsilon_{ijk}\,\frac{\partial}{\partial x_j}\left(\varepsilon_{kmn}\,\frac{\partial u_n}{\partial x_m}\right) =$$

$$\cancel{(\nabla u)_{ij} = \frac{\partial u_i}{\partial x_j}}$$

$$\cancel{(\nabla^T u)_{ij} = \frac{\partial u_j}{\partial x_i}}$$

Ver abajo *

$$= \varepsilon_{ijk}\,\varepsilon_{kmn}\,\frac{\partial^2 u_n}{\partial x_j\,\partial x_m} = \varepsilon_{kij}\,\varepsilon_{kmn}\,\frac{\partial^2 u_n}{\partial x_j\,\partial x_m} =$$

$$= (\delta_{im}\,\delta_{jn} - \delta_{in}\,\delta_{jm})\,\frac{\partial^2 u_n}{\partial x_j\,\partial x_m} =$$

$$(\nabla^T u)_{ij}$$

$$\cancel{\frac{\partial}{\partial x_j}\left(\frac{\partial u_j}{\partial x_i}\right) = \nabla\cdot\nabla^T u}$$

$$\cancel{= \frac{\partial}{\partial x_j}(\nabla^T u)_{ij}}$$

$$\cancel{= \frac{\partial}{\partial x_j}\left(\frac{\partial u_j}{\partial x_i}\right)}$$

$$= \frac{\partial^2 u_j}{\partial x_j\,\partial x_i} - \frac{\partial^2 u_i}{\partial x_j\,\partial x_j} = \frac{\partial}{\partial x_i}\left(\overbrace{\frac{\partial u_j}{\partial x_j}}^{\nabla\cdot u}\right) - (\Delta u)_i$$

$$= [\nabla(\nabla u)]_i - (\Delta u)_i \Rightarrow \nabla\times(\nabla\times u) = \nabla(\nabla\cdot u) - \Delta u$$

$$\frac{\partial}{\partial x_i}\left(\frac{\partial u_j}{\partial x_j}\right) = \frac{\partial}{\partial x_j}\left(\frac{\partial u_j}{\partial x_i}\right) = \frac{\partial}{\partial x_j}(\underline{\nabla}\,\underline{u})_{ji} = \frac{\partial}{\partial x_j}\left[(\underline{\nabla}\,\underline{u})^T\right]_{ij} =$$

$$= \left[\underline{\nabla}\cdot(\underline{\nabla}\,\underline{u})^T\right]_i$$

$$\left[\underline{\nabla}\times(\underline{\nabla}\times\underline{u})\right]_i = \left[\underline{\nabla}\cdot(\underline{\nabla}\,\underline{u})^T\right]_i - (\underline{\Delta}\,\underline{u})_i \implies$$

$$\Rightarrow \underline{\nabla}\times(\underline{\nabla}\times\underline{u}) = \underline{\nabla}\cdot(\underline{\nabla}\,\underline{u})^T - \underline{\Delta}\,\underline{u}$$

$$\underline{\underline{T}}^T \Rightarrow \begin{cases}(T_{ij})^T = T_{ji}\\ (\underline{\underline{T}}^T)_{ij} = T_{ji}\end{cases}$$

$$\text{si } \underline{\underline{T}} = \underline{\nabla}\,\underline{u} \Rightarrow T_{ij} = \frac{\partial u_i}{\partial x_j} \quad \wedge \quad \text{si } \underline{\underline{T}} = (\overbrace{\underline{\nabla}\,\underline{u}}^{\underline{\underline{T}}})^T \Rightarrow (\underline{\underline{T}})_{ij} = \left(\frac{\partial u_i}{\partial x_j}\right)^T = \frac{\partial u_j}{\partial x_i}$$

$$\text{si } \underline{\nabla}\cdot\underline{\underline{T}} \Rightarrow (\nabla\cdot\underline{\underline{T}})_i = \frac{\partial T_{ij}}{\partial x_j} \quad \wedge \quad \text{si } \underline{\nabla}\cdot\underline{\underline{T}}^T \Rightarrow (\underline{\nabla}\cdot\underline{\underline{T}}^T)_i = \frac{\partial}{\partial x_j}(\underline{\underline{T}}^T)_{ij} = \frac{\partial T_{ji}}{\partial x_j}$$

h. $\nabla \times (\mathbf{u} \times \mathbf{v}) = (\nabla \bullet \mathbf{v})\mathbf{u} - (\nabla \bullet \mathbf{u})\mathbf{v} + (\mathbf{v} \bullet \nabla)\mathbf{u} - (\mathbf{u} \bullet \nabla)\mathbf{v}$

$$\left[\underline{\nabla} \times (\underline{u} \times \underline{v})\right]_i = \varepsilon_{ijk} \cdot \frac{\partial}{\partial x_j}\left(\varepsilon_{kmn}\, u_m v_n\right) =$$

$$= \underset{\varepsilon_{kij}}{\varepsilon_{ijk}} \cdot \varepsilon_{kmn} \cdot \frac{\partial}{\partial x_j}(u_m . v_n) =$$

$$= (\delta_{im} . \delta_{jn} - \delta_{in}\, \delta_{jm}) . \left(\frac{\partial}{\partial x_j}(u_m) . v_n + u_m . \frac{\partial v_n}{\partial x_j}\right)$$

$$= \frac{\partial u_i}{\partial x_j} v_j - \frac{\partial u_j}{\partial x_j} . v_i + u_i . \frac{\partial v_j}{\partial x_j} - u_j . \frac{\partial v_i}{\partial x_j}$$

$$u_m . \frac{\partial v_i}{\partial x_m}$$

Forma 1

$$= \left[\underset{A_{ij}}{(\underline{\nabla}\, \underline{u})} \cdot \underset{v_j}{\underline{v}}\right]_i - \left[(\underline{\nabla} \cdot \underline{u})\, \underline{v}\right]_i + \left[\underline{u}\, (\underline{\nabla} \cdot \underline{v})\right]_i - \left[\underset{A_{ij}}{(\underline{\nabla}\, \underline{v})} \cdot \underset{u_j}{\underline{u}}\right]_i$$

$$\underline{\nabla} \times (\underline{u} \times \underline{v}) = \underset{\text{tensor 2do orden}}{(\underline{\nabla}\, \underline{u})} \cdot \underline{v} - \underset{\text{escalar}}{(\underline{\nabla} \cdot \underline{u})}\, \underline{v} + \underset{\text{escalar}}{(\underline{\nabla} \cdot \underline{v})}\, \underline{u} - \underset{\text{tensor de 2do orden}}{(\underline{\nabla}\, \underline{v})} \cdot \underline{u}$$

Forma 2 $\quad \left(v_k \frac{\partial}{\partial x_k}\right) u_j$

$$= (\underline{v} \cdot \underline{\nabla})[\underline{u}]_i - (\underline{\nabla} \cdot \underline{u})[\underline{v}]_i + (\underline{\nabla} \cdot \underline{v})[\underline{u}]_i - (\underline{u} \cdot \underline{\nabla})[\underline{v}]_i$$

$$\underline{\nabla} \times (\underline{u} \times \underline{v}) = (\underline{v} \cdot \underline{\nabla})\, \underline{u} - (\underline{\nabla} \cdot \underline{u})\, \underline{v} + (\underline{\nabla} \cdot \underline{v})\, \underline{u} - (\underline{u} \cdot \underline{\nabla})\, \underline{v}$$